Low-Light Dense Monocular SLAM on Jetson AGX Orin

Aditya Abhang · with Mihir Chauhan
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Abstract. Feature-based visual SLAM degrades sharply in extreme low light: at ISO 102400 there is not enough photon signal for repeatable keypoints, and a stock matcher spends the majority of a sequence in relocalization. This project takes a learned *dense* SLAM system — MASt3R-SLAM, paired with Dark3R, a low-light LoRA fine-tune of its neural matcher — and makes it run in real time on an NVIDIA Jetson AGX Orin, the class of compute a small aerial platform can carry, while quantifying how trajectory accuracy degrades as a function of scene noise and exposure. Contributions: an embedded deployment that resolves the unified-memory CUDA, allocator, and inter-process constraints of the AGX; an exposure-confidence preprocessing stage that lifts only the frames it is confident are under-exposed; a raw-domain sensor-noise degradation dataset spanning +3 dB to −10 dB SNR with a power-matched white-noise control; and a metric evaluation harness reporting ATE, RPE, scale drift and map statistics against clean-image references. On 500 frames of real ISO-102400 footage the low-light-tuned system reaches 1.27 m ATE at 4.75 FPS on device, versus 2.35 m and ~55% of frames lost to relocalization for the stock model.

1 Motivation

Autonomous flight in caves, tunnels, collapsed structures and night-time forest canopy puts a monocular camera into a regime classical SLAM was not built for. ORB- and photometric-based front ends need image gradients that survive from frame to frame; at very high ISO the dominant signal is sensor noise, which is spatially white and temporally uncorrelated, so corner detectors fire on noise and descriptor matches do not repeat. Learned dense matchers such as MASt3R regress a pointmap directly from an image pair and are far more robust to appearance change, and Dark3R fine-tunes that matcher specifically for low light. The open problem for a drone is that this robustness comes from a 689 M-parameter transformer that assumes a desktop-class discrete GPU. This report covers making that stack usable on the Jetson AGX Orin and measuring how far into the dark it holds a trajectory.

2 System overview

MASt3R-SLAM runs a per-frame tracking front end (a 7-DoF Gauss–Newton pose solve against the current keyframe's pointmap) and a backend that maintains a global factor graph with retrieval-based loop closure. Dark3R ships as LoRA adapters over the MASt3R ViT-L base; merging the adapters back into the base weights produces a plain checkpoint the SLAM stack consumes without code changes. In the deployment configuration the backend runs as a thread sharing one model instance and one semaphore with the front end, and global optimisation is serialized behind each keyframe insertion, which makes runs deterministic and bit-reproducible — the property that makes controlled A/B evaluation possible.

3 Contributions

3.1 Jetson AGX Orin bring-up

The stock stack assumes discrete-GPU semantics. On the AGX's 64 GB unified memory, spawning the backend as a separate process crashes while mapping CUDA IPC handles (`pidfd_getfd: operation not permitted`); the desktop's `expandable_segments` allocator relies on virtual-memory APIs the integrated GPU does not expose, producing an immediate out-of-memory with tens of gigabytes free; and the older on-device Torch build lacks `mmap` checkpoint loading, so the checkpoint was read twice and startup stalled for minutes. Resolved by running the backend as a thread, disabling expandable segments on the iGPU, de-duplicating the checkpoint read, and adding a CPU Cholesky fallback for the pose solve when the on-device factorization fails. Retrieval is kept on CPU on the AGX; on GPU it is ~25× faster but the extra ~350 MB of device memory contributed to intermittent cuDNN execution failures.

3.2 Dark3R LoRA merge and validation

The LoRA-to-base merge was validated against both the stock model and the adapter-applied model to confirm the merge itself introduces no regression before it entered the deployment builds (Phase 2 validation).

3.3 Exposure-confidence preprocessing

A per-frame estimate of under-exposure from the 90th-percentile luma `L`. An exposure confidence `c = clip(L / L*, 0, 1)` is formed against a target `L* = 0.5`; frames with `c ≥ 0.9` pass through bit-for-bit unchanged. Below that, a gain toward `L*` is applied, capped at 8× and scaled by `(1 − c/0.9)` so the correction fades out as confidence rises. Unlike a global mean/standard-deviation normalization, which rewrites every frame, this never touches a well-exposed frame and does not break photometric consistency between neighbours. It is gated by a config flag so baseline runs are unaffected.

3.4 Raw-domain SNR degradation dataset

To measure noise sensitivity independently of scene content, a per-photosite Poisson–Gaussian sensor-noise model is applied to normalized linear CFA data and demosaiced back through LibRaw. A calibration loop tunes the noise scale until each level lands within 0.4 dB of target, giving seven matched levels from +3 dB to −10 dB (310 frames each; the −10 dB level measures −9.60 ± 1.33 dB). A power-matched additive-white-Gaussian variant (−9.58 dB) isolates how much of the tracking loss is caused by the demosaic's spatial and chroma correlation rather than white noise alone.

3.5 Evaluation harness

Absolute and relative pose error (computed with `evo`), plus keyframe count, map-point count, Sim(3) scale drift, relocalization count and on-device frame rate. For synthetic degradations the reference is the clean-image SLAM trajectory; for real ISO-102400 footage it is Dark3R's COLMAP long-exposure poses.

4 Results

4.1 Real low light — 500 frames, ISO 102400

model	ATE RMSE (m)	keyframes	map pts	reloc failures	FPS
stock MASt3R	2.350	34	6.1 M	273 / 274	2.45
Dark3R merged	**1.265**	21	3.8 M	58 / 60	4.75

Ground truth = COLMAP long-exposure poses. The stock model spends ~55% of frames in relocalization; the low-light-tuned model cuts ATE by 46% and roughly doubles on-device frame rate.

4.2 Synthetic darkening — TUM fr1, Sim(3)-aligned ATE (m)

model	params	clean ATE	dark ATE	FPS
stock MASt3R	689 M	0.1186	0.1050	5.06
Dark3R merged	689 M	**0.0851**	**0.1035**	4.99

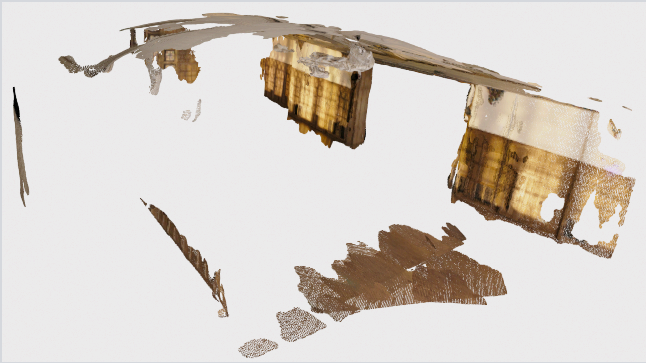
The low-light LoRA improves general matching, not only low light: its largest gain is a 28% ATE reduction on the clean sequence.

4.3 Paired-burst study — 47 fixed viewpoints, four capture conditions

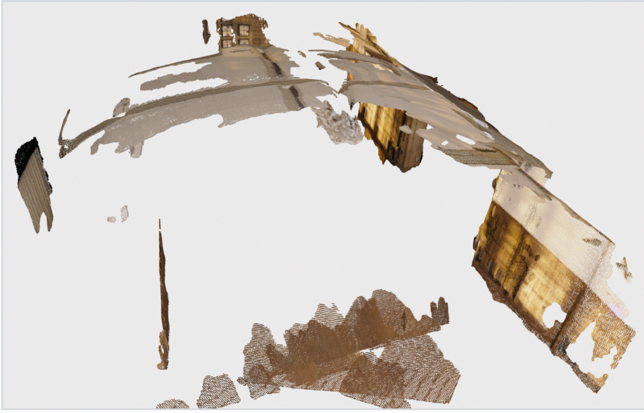
A single 47-position indoor path was captured four ways: a clean long exposure, two short noisy exposures, and a motion-blurred frame per position. Each was reconstructed independently; the clean run is the reference trajectory (the four sequences trace the same physical path).

run	mean luma	keyframes	map pts	ATE RMSE (m)	RPE rot (°/step)	Sim(3) scale	reloc
clean (reference)	0.09–0.26	16	550 k	—	—	1.000	0
noisy #0	~0.03	22	758 k	0.156	0.79	0.952	0
noisy #1	~0.007	23	792 k	0.316	1.22	0.907	0
motion blur	0.20–0.27	21	723 k	0.203	0.69	1.069	0

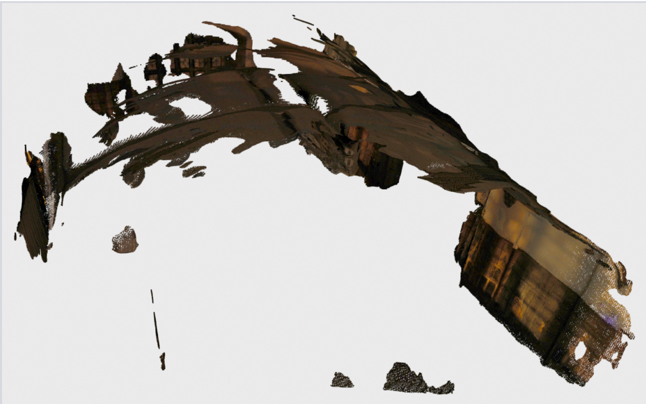
Path length ~12 m. All four sequences tracked end to end with zero relocalization and zero Cholesky failures. ATE grows from 1.3% of path length at mean luma 0.03 to 2.7% at mean luma 0.007, with a 9% depth-scale contraction in the darkest case; motion blur adds absolute drift but the lowest per-step rotation error, i.e. smooth bias rather than jitter.



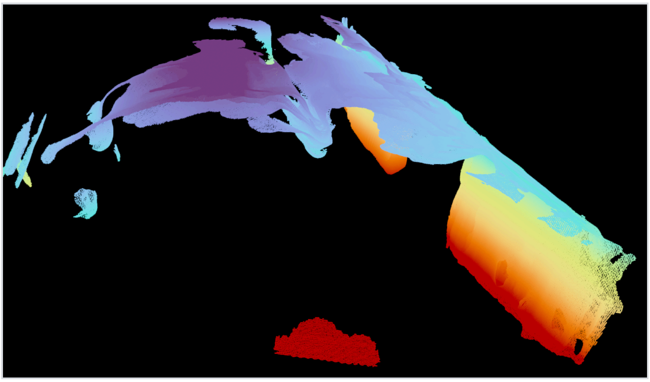
Clean long-exposure reconstruction — windows, panelled walls, parquet floor and a column recovered as a dense coloured point cloud.



Same reconstruction, corridor view along the 47-position path.



Short-exposure noisy input (mean luma ~0.03): the same geometry survives, darker and with more spurious points baked in near the walls.



Darkest input (mean luma ~0.007). RGB is essentially black, so points are coloured by height — the room's walls, floor and doorway pillar are still resolved.

4.4 Degradation signature

Across both the SNR sweep and the paired-burst study, lower SNR *raises* keyframe and point counts rather than lowering them: the matcher's match fraction drops, the tracker inserts keyframes more aggressively, and sensor noise is fused into the map as spurious geometry. The noisiest paired-burst runs produced ~40% more keyframes and points than the clean run. A denser-looking cloud is therefore a degradation symptom, not a better map, and the evaluation weights trajectory error over raw map size. Under-exposure and noise are also separable axes: the SNR-sweep frames are digitally re-gained during synthesis and sit at normal brightness, so the exposure-confidence stage is correctly a no-op on them and its effect must be demonstrated on a separately de-gained variant.

5 Limitations and future work

- On-device frame rate (1–5 FPS depending on sequence and keyframe density) is below the ~20 FPS a fast platform wants; the AGX is dispatch-bound on the neural path. TensorRT export (traceable RoPE, shallow-decoder heads) and CUDA-graphed inference are the next levers.
- Ground truth for real low light is COLMAP on long exposures of the same scene, not an independent motion-capture track; absolute ATE values inherit that reference's error.

- The paired-burst dataset is 47 discrete viewpoints, not a continuous handheld pass, so it stresses wide-baseline matching more than temporal tracking.
- Target hardware is ultimately the Orin Nano 8 GB, which will require the distilled student model (a parallel effort by M. Chauhan, out of scope here).

6 Conclusion

A learned dense SLAM system, tuned for low light and adapted to the Jetson AGX Orin's unified-memory constraints, tracks real ISO-102400 footage end to end at 4.75 FPS with 46% lower trajectory error than the stock model, where the stock model fails outright. A raw-domain degradation dataset and a metric evaluation harness make the failure onset measurable: trajectory error stays within ~3% of path length down to mean scene luma below 0.01, and the dominant failure mode at low SNR is spurious map geometry from fused sensor noise rather than lost tracking.

References

1. R. Murai, E. Dexheimer, A. J. Davison. *MASt3R-SLAM: Real-Time Dense SLAM with 3D Reconstruction Priors.* arXiv:2412.12392.
2. A. Guo et al. *Dark3R: Low-Light Monocular Geometry Estimation.* github.com/andrewyguo/Dark3R.
3. H. Lu et al. *EC3R-SLAM: Efficient and Consistent Monocular Dense SLAM.* arXiv:2510.02080.
4. M. Grupp. *evo: Python package for the evaluation of odometry and SLAM.* github.com/MichaelGrupp/evo.

Reconstructions rendered offscreen with Open3D on the Jetson. Measured figures in §4.1–4.2 are from the project's shared evaluation logs; §4.3 was produced for this report.